



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

AIOT

13-05-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	60.00	1.06	50.00	7.00	2.00	4.60	3.21 to 3.25	159.00 to 159.30	10.00	0.50
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	3	1	3	4	3	5	3	3		

PART-2 : CHEMISTRY

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	95.52	1.60	200.00	0.40	266.80	56.16 or 54.17	4.80	12.00	60.00	7.64
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	5	5	4	7	5	6	9	5		

PART-3 : MATHEMATICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	144.00	441.00	100.00	1211.00	30.00	60.00	771.00	125.00	15.00	18.00
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	3	3	1	4	8	7	3	3		

HINT – SHEET

PART-1 : PHYSICS

SECTION-II

1. Ans (60.00)

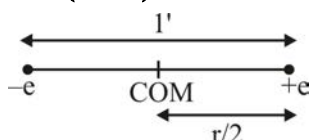
Time taken by sound reach the ground

$$= \frac{20000}{340} = \frac{1000}{17} \text{ s}$$

Distance moved by plane in that time

$$= 1020 \times \frac{1000}{17} = 60 \text{ km}$$

2. Ans (1.06)



$$\frac{\text{K. e. e.}}{r^2} = \frac{mv^2}{r/2} \quad \dots (1)$$

$$mvr = \frac{nh}{2\pi} \quad \dots (2)$$

From 1 and 2, we get

$$r = \frac{2n^2 h^2}{4\pi^2 m k e^2}$$

$$\therefore n = 2$$

3. Ans (50.00)

The capacitors $2\mu\text{F}$ and $1\mu\text{F}$ are in parallel combination.

So, let $1\mu\text{F}$ get a charge q_1 and $2\mu\text{F}$ get a charge of q_2 .

$$q_1 : q_2 = C_1 : C_2 = 1 : 2$$

$$\text{So, given } q_2 = 20 \mu\text{J}$$

$$\text{So, } q_1 = 10 \mu\text{J}$$

$$\text{So, energy stored in } C_1 = \frac{10^2}{2 \times 1} = 50 \mu\text{J}$$

$$\text{and energy stored in } C_2 = \frac{20^2}{2 \times 2} = 100 \mu\text{J}$$

So, total energy stored in capacitor

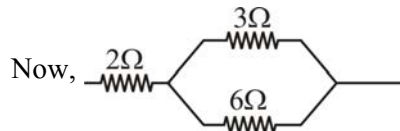
$$= 100 + 50 = 150 \mu\text{J}$$

Now, if we calculate energy dissipated in resistors from the formulation

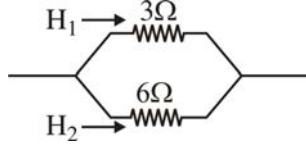
$$H = \int i^2 R \, dt$$

It will be same as E_{stored} in capacitor

$$\text{So, } H_{\text{Total}} = 150 \mu\text{J}$$



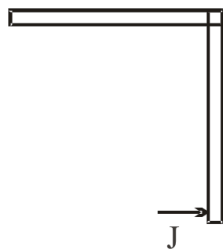
Energy dissipated in the branch of 3Ω & 6Ω will be $75\mu\text{J}$.



$$H_1 : H_2 = 6 : 3 = 2 : 1$$

$$\text{So, } H_1 = \frac{2}{3} \times 75 = 50\mu\text{J}$$

4. **Ans (7.00)**



$$J - J_1 = mv_1$$

$$J_1 = mv_2$$

$$\frac{J\ell}{2} + J_1 \frac{\ell}{2} = \frac{m\ell^2}{12} \omega$$

$$J + J_1 = \frac{m\ell}{6} \omega$$

$$v_2 = v_1 - \frac{\ell}{2} \omega^2$$

$$\frac{J_1}{m} = \frac{J}{m} - \frac{J_1}{m} - \frac{3}{m} (J + J_1)$$

$$J_1 = -2J - 4J_1$$

$$5J_1 = -2J$$

$$J_1 = \frac{-2}{5} J$$

$$\frac{7}{5} J = mv_1 \Rightarrow v_1 = \frac{7J}{5m} = \frac{7 \times 35}{5 \times 7} = 7 \text{ m/s}$$

5. **Ans (2.00)**

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 10^{-3} \times 80 \times 10^{-6}}}$$

$$\omega_0 = \frac{1}{\sqrt{4000 \times 10^{-9}}} = 500 \text{ rad/s}$$

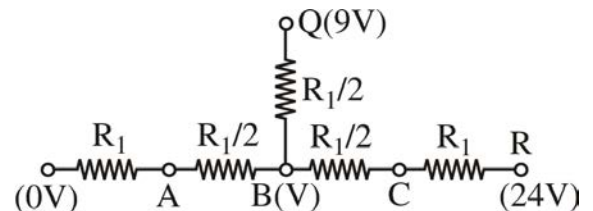
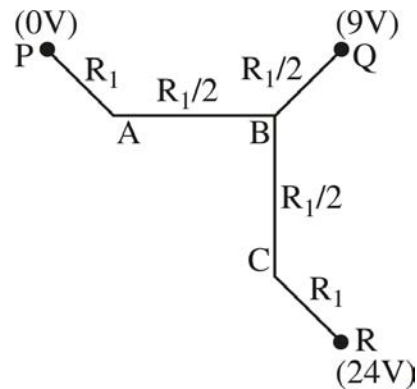
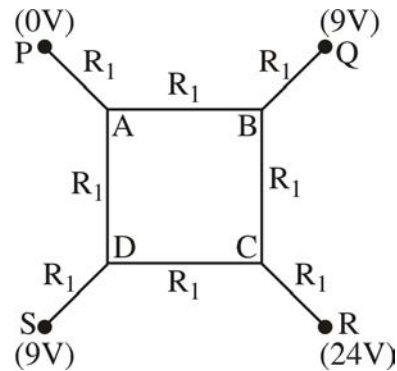
$$Q = \frac{\omega_0 L}{R} = \frac{500 \times 50 \times 10^{-3}}{12.5}$$

$$Q = \frac{25}{12.5} = 2$$

6. **Ans (4.60)**

$$\text{Resistance of are arm } R_1 = \frac{R}{8} + \frac{R}{16} + \dots$$

$$R_1 = \frac{R}{8} + \left(1 + \frac{1}{2} + \dots\right) = \frac{R}{4}$$



$$\frac{V-9}{R_1/2} + \frac{V-24}{3R_1/2} + \frac{V-0}{3R_1/2} = 0$$

$$3(V-9) + (V-24) + (V-0) = 0$$

$$5V = 27 + 24 = 51$$

$$V = 10.2 \text{ V}$$

$$V_{CB} = \frac{R_1/2}{R_1/2 + R_1} \times (24 - 10.2)$$

$$= 8 - 3.4 = 4.6\text{V}$$

7. **Ans (3.21 to 3.25)**

$$N^{13} \rightarrow C^{13} + \beta^+ + \nu + 1.2 \text{ MeV (M : Atomic masses)}$$

$$(M_N - M_C - 2m_e)c^2 = 1.2 \text{ MeV}$$

$$P + C^{13} \rightarrow N^{13} + n$$

$$Q = (m_P - m_C - m_N - m_n)c^2 \text{ (m : Nuclear masses)}$$

added and subtracted $7m_e$ to get Q value in atomic masses.

$$Q = (M_H - M_C - M_N - M_n) = -[(M_n - M_H) + (M_N - M_C)]$$

$$= -[0.78 + 1.2 + 2 \text{ me}] = -3 \text{ MeV}$$

$$E_{th} = |Q| \left(1 + \frac{m_p}{m_c} \right)$$

$$= 3 \times \left(1 + \frac{1}{13} \right) = 3.23 \text{ MeV}$$

8. **Ans (159.00 to 159.30)**

$$V_C = \frac{V}{Z}x - jx_C; Z = R - jx_C$$

$$V_R = \frac{V}{Z}xR$$

$$V_{AB} = V_R - V_C = \frac{V}{Z}(R + jx_C)$$

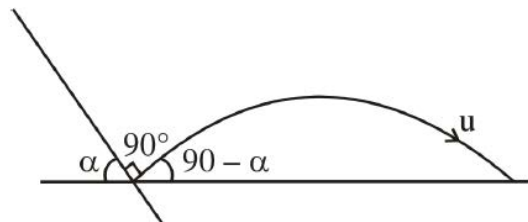
$$= V \left(\frac{R + jx_C}{R - jx_C} \right) = V \left(\frac{R^2 + 2jx_C R - x_C^2}{R^2 + x_C^2} \right)$$

$$\Delta\phi = \frac{\pi}{2} \Rightarrow R^2 - x_C^2 = 0$$

$$\omega = \frac{1}{R_C} = \frac{1}{10^5 \times 10^{-11}} = 10^6 \text{ rad/s}$$

$$f = \frac{1000}{2\pi} = 159.15 \text{ kHz}$$

10. **Ans (0.50)**



$$L = \frac{u^2 \sin 2(90 - \alpha)}{g}$$

$$l = \frac{v^2 \sin 2(90 - \alpha)}{g}$$

$$\sqrt{\frac{l}{L}} = \frac{v}{u} = e = 0.50$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (3)**

$$h = \frac{2T \cos 45^\circ}{rpg} \quad \dots(i)$$

$$\frac{h}{\sqrt{2}} = \frac{2T \cos \theta}{rpg} \quad \dots(ii)$$

$$(i) \& (ii) \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

4. **Ans (4)**

Total momentum (horizontal) of the water filled inside the tank till time t .

$$P = 2\lambda tv_0$$

Using linear momentum conservation

$$v = \frac{2\lambda tv_0}{M + 2\lambda t}$$

$$v = \frac{2 \times 2 \times 100 \times 5}{100 + 2 \times 2 \times 100} = 4 \text{ m/s}$$

6. **Ans (5)**

$$\frac{dq}{dt} = I - I_d; I_d = \frac{q}{CR}$$

$$E - IR - \frac{q}{C} = 0$$

$$E - \left(\frac{q}{RC} + \frac{dq}{dt} \right) R - \frac{q}{C} = 0$$

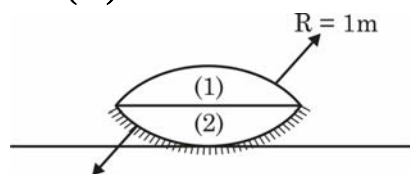
$$\int_0^q \frac{dq}{CE - q} = \int_0^t \frac{dt}{RC}$$

$$q = \frac{CE}{2} \left(1 - e^{-2t/RC} \right)$$

at $t = \infty$

$$q = \frac{CE}{2}$$

7. Ans (3)



$$\frac{1}{f_1} = (1.5 - 1) \left(\frac{1}{1} \right) = 0.5 \text{ m}^{-1}$$

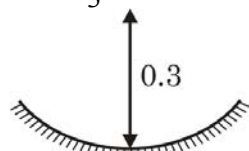
$$\frac{1}{f_2} = \left(\frac{4}{3} - 1 \right) \left(\frac{1}{0.5} \right) = \frac{1}{3} \times \frac{1}{0.5} = \frac{2}{3} \text{ m}^{-1}$$

Power of the system

$$P = 2P_1 + 2P_2P_m$$

$$= 2 \times 0.5 + 2 \times \frac{2}{3} + \frac{1}{0.25}$$

$$= 1 + \frac{4}{3} + 4 = \frac{3 + 4 + 12}{3} = \frac{19}{3}$$



Focal length of equivalent mirror = $-\frac{3}{19} \text{ m}$

$$\frac{1}{v} + \frac{1}{-0.3} = \frac{1}{-\frac{3}{19}}$$

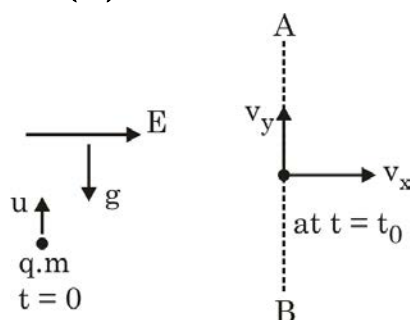
$$\frac{1}{v} = \frac{1}{-0.3} - \frac{19}{-3} = \frac{10}{3} - \frac{19}{3} = \frac{-9}{3} = -3$$

$$v = -\frac{1}{3} \text{ m}$$

$$\Rightarrow d = -\frac{3}{10} + \frac{1}{3} = \frac{1}{30}$$

$$\Rightarrow \frac{1}{30} = \frac{x}{90} \Rightarrow x = 3 \text{ m}$$

8. Ans (3)



$$t \leq t_0; v_x = \frac{qE}{m} t_0 = gt_0$$

$$v_y = u - gt_0$$

just after AB, $\vec{V} = \text{constant} \Rightarrow \vec{F}_{\text{net}} = 0$

$$\Rightarrow q\vec{E} + q(\vec{v} \times \vec{B}) + m\vec{g} = 0$$

$$\Rightarrow qE\vec{i} + qv_x B\vec{j} - qv_y B\vec{i} - mg\vec{j} = 0$$

$$\Rightarrow E = B(u - gt_0) \text{ and } qBt_0 = m$$

$$\Rightarrow u = 2gt_0 = 3 \text{ m/s}$$

PART-2 : CHEMISTRY

SECTION-II

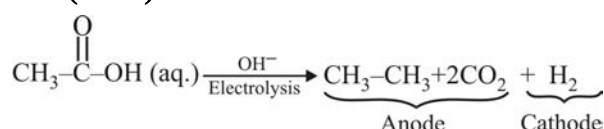
1. Ans (95.52)

$$\%S_{N2} = \frac{(3.2 \times 10^{-5} \times 1) \times 100}{(3.2 \times 10^{-5} + 1.5 \times 10^{-6})}$$

$$= \frac{32}{32 + 1.5} \times 100$$

$$= 95.52\%$$

2. Ans (1.60)



$$n_{\text{CO}_2} = 2 \times n_{\text{CH}_3\text{CH}_3} \quad n_{\text{CO}_2} = \frac{4.4}{44} = .1$$

$$n_{\text{H}_2} = \frac{1}{2} \times n_{\text{CO}_2} \quad n_{\text{CH}_3-\text{CH}_3} = \frac{1}{2} \times .1$$

$$= .05$$

$$n_{\text{H}_2} = \frac{1}{2} \times .1$$

$$= .05$$

$$W_{\text{C}_2\text{H}_6} + W_{\text{H}_2} = .05 \times 30 + 0.05 \times 2$$

$$= 1.5 + .1 = 1.6 \text{ gm}$$

3. Ans (200.00)

Density of gas will remain constant as its mole and volume remains unchanged.

$$\Delta u_{\text{rms}} = u_2 - u_1 = \sqrt{\frac{3P_2}{d}} - \sqrt{\frac{3P_1}{d}}$$

$$= \sqrt{\frac{3 \times 9 \times 10^4}{0.75}} - \sqrt{\frac{3 \times 4 \times 10^4}{0.75}} = 200 \text{ m/s.}$$

4. Ans (0.40)

$$K = G \cdot G^*$$

$$\therefore \frac{K_{\text{NH}_4\text{OH}}}{K_{\text{KCl}}} = \frac{\frac{1}{10^5} \times G^*}{\frac{1}{240} \times G^*}$$

$$\text{or, } \frac{K_{\text{NH}_4\text{OH}}}{0.2714} = \frac{240}{10^5} \Rightarrow K_{\text{NH}_4\text{OH}} = \frac{0.2714 \times 240}{10^5} \text{ Sm}^{-1}$$

$$\text{Now, for } \text{NH}_4\text{OH}, \alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{K/C}{\Lambda_m^0}$$

$$= \frac{\frac{0.2714 \times 240}{10^5}}{(6 \times 10^{-5} \times 10^3) \times [7.34 \times 10^{-3} + 1.98 \times 10^{-2}]}$$

$$= 0.40$$

5. **Ans (266.80)**

$$P^{1-\gamma} \cdot T^{\gamma} = \text{constant}$$

$$\therefore T_2 = T_1 \cdot \left(\frac{P_1}{P_2} \right)^{\frac{29.88-8.3}{29.88}-1}$$

$$= 266.80 \text{ K}$$

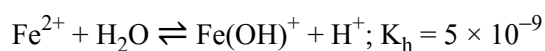
6. **Ans (56.16 or 54.17)**

$$P = x_2 \cdot P^{\circ}$$

$$30 = \frac{\frac{1000}{18}}{\frac{x}{58.5} \times 2 + \frac{1000}{18}} \times 31$$

$$\Rightarrow x = 54.167$$

7. **Ans (4.80)**



$$(0.05-x)_M \quad xM \quad xM$$

$$\approx 0.05M$$

$$\text{Now, } 5 \times 10^{-9} = \frac{x \cdot x}{0.05} \Rightarrow x = (25 \times 10^{-11})^{1/2}$$

$$\text{And, pH} = -\log x = 4.80$$

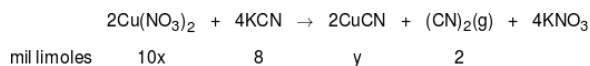
10. **Ans (7.64)**

$$x = 0.4$$

$$y = 4$$

$$z = 0.24$$

$$p = 3$$



$$\text{mil limoles} \quad 10x \quad 8 \quad y \quad 2$$

$$10x = 4 \Rightarrow x = 0.4 \dots(i)$$

$$y = 4 \dots(ii)$$

Also all KCN was consumed in this step.



$$\begin{array}{ccc} \text{Re action} & & \\ \text{mmoles} & 4 & 12 \end{array}$$

$$\therefore 50z = 12 \Rightarrow z = 0.24 \dots(iii)$$

O.S. of C in $(\text{CN})_2$ is +3

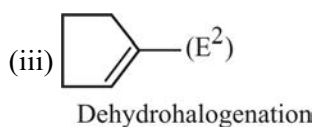
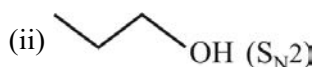
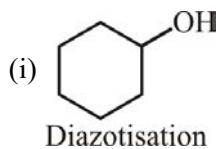
$$\therefore p = 3 \dots(iv)$$

PART-2 : CHEMISTRY

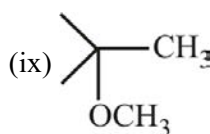
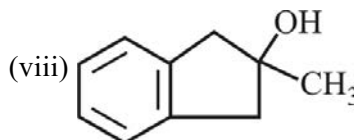
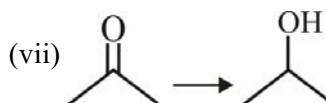
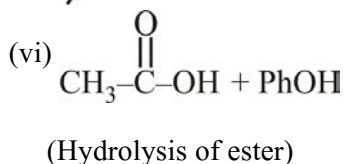
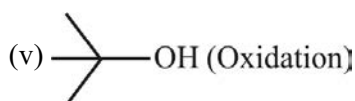
SECTION-III

1. **Ans (5)**

(i), (ii), (v), (vii), (viii)

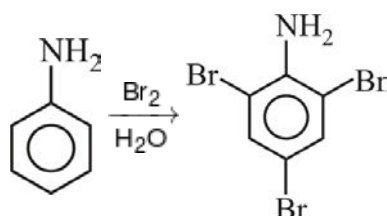
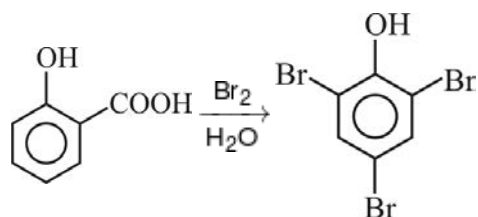
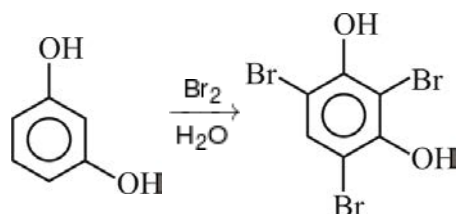
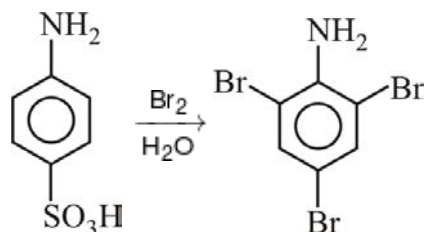
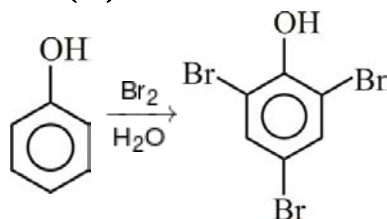


(iv) No reaction



(electrophilic addition)

2. Ans (5)



3. Ans (4)

$$\frac{yz}{108x}$$

$$x = 7, y = 24, z = 7 \times 18 = 126$$

Ans. 4

4. Ans (7)

$$n_{\text{NH}_3} = .1 \times .2 = .02 \quad W_{\text{CH}_4} = .04 \times .5 \times 16$$

$$= n_{\text{HCl}}$$

$$n_{\text{P}} = .04$$

$$W_{\text{H}_2} = .04 \times \frac{1}{2} \times 2$$

$$W_{\text{NH}_3} = .04 \times .5 \times 17$$

5. Ans (5)

NCERT Based.

B, E, G, H and I are correct.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (144.00)

$$g\left(\frac{\pi}{12}\right) = \int_{\pi/12}^{\pi/4} \cot^{-1}(2 - \tan x) dx = x \cot^{-1}(2 - \tan x) \Big|_{\pi/12}^{\pi/4}$$

$$- \int_{\pi/12}^{\pi/4} \frac{x \sec^2 x}{1 + (2 - \tan x)^2} dx = \left(\frac{\pi}{4} \times \frac{\pi}{4} - \frac{\pi}{12} \times \frac{\pi}{6} \right) - I$$

In I ; put $2 - \tan x = \tan \theta$ then,

$$g\left(\frac{\pi}{12}\right) = \frac{7\pi^2}{144} - \int_{\pi/4}^{\pi/3} \left(\frac{\pi}{2} - \cot^{-1}(2 - \tan \theta) \right) d\theta = \frac{\pi^2}{144} + g\left(\frac{\pi}{3}\right)$$

2. Ans (441.00)

$$D = -a, \quad D_1 = -2(a - b + 6)$$

$$D_2 = -2a - b + 6, \quad D_3 = 6 - b$$

for unique solution, $b \neq 0$

$$\therefore L = 20 \times 21 = 420$$

for no solution, $a = 0, b \neq 6,$

$$\therefore M = 1 \times 20 = 20$$

for infinitely many solutions

$$a = 0, b = 6, N = 1 \times 1 = 1$$

3. Ans (100.00)

$$|2z_1 + \bar{z}_2|^2 - |1 + 2z_1 z_2|^2 = 8 - 9$$

$$\Rightarrow 4|z_1|^2 + |z_2|^2 - 1 - 4|z_1|^2|z_2|^2 = -1$$

$$\Rightarrow \frac{4}{|z_2|^2} + \frac{1}{|z_1|^2} = 4$$

$$A.M. \geq H.M.$$

$$\frac{|z_1|^2 + 4|z_2|^2}{5} \geq \frac{5}{\frac{1}{|z_1|^2} + \frac{4}{|z_2|^2}}$$

$$\Rightarrow |z_1|^2 + 4|z_2|^2 \geq \frac{25}{4}$$

4. Ans (1211.00)

$$b = \sum_{r=1}^{11} \cos^2 \left(\frac{r\pi}{24} \right) = \sum_{r=1}^{11} \sin^2 \left(\frac{\pi}{2} - \frac{r\pi}{24} \right)$$

$$= \sum_{r=1}^{11} \sin^2 \left(\frac{(12-r)\pi}{24} \right) = \sum_{r=1}^{11} \sin^2 \left(\frac{r\pi}{24} \right) = a$$

$$\text{Similarly } d = \sum_{r=1}^{11} \cot^2 \left(\frac{r\pi}{24} \right)$$

$$= \sum_{r=1}^{11} \tan^2 \left(\frac{\pi}{2} - \frac{r\pi}{24} \right)$$

$$= \sum_{r=1}^{11} \tan^2 \left(\frac{(12-r)\pi}{24} \right) = \sum_{r=1}^{11} \tan^2 \left(\frac{r\pi}{24} \right) = c$$

$$\text{Also } a + b = \sum_{r=1}^{11} \left(\sin^2 \frac{r\pi}{24} + \cos^2 \frac{r\pi}{24} \right)$$

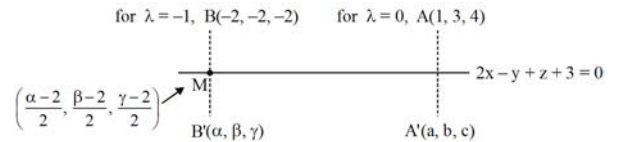
$$= \sum_{r=1}^{11} 1 = 11 \text{ and } \log_{(2d-c)}(2c-d) = \log_d c = 1$$

$$a^3 + b^3 - 11(a+b) + 33ab = a^3 + b^3 - (a+b)^2$$

$$+ 3(a+b)(ab) = (a+b)^3 - (a+b)^2 = 11^3 - 11^2$$

$$= 1210$$

5. Ans (30.00)



$$\text{Point M satisfy plane } \Rightarrow 2\alpha - \beta + \gamma + 6 = 0 \dots(i)$$

$$\text{and } \frac{\alpha+2}{2} = \frac{\beta+2}{-1} = \frac{\gamma-2}{1} = \lambda \dots(ii)$$

$$\Rightarrow \text{From (i) and (ii) we get, } \lambda = \frac{-6}{7}$$

$$\therefore (\alpha, \beta, \gamma) \equiv \left(\frac{-26}{7}, \frac{-8}{7}, \frac{8}{7} \right)$$

$$\text{Similarly } A'(-3, 5, 2)$$

$$\Rightarrow \text{Equation of line L passes through point A' and}$$

$$B' \text{ is}$$

$$\frac{x+3}{5} = \frac{y-5}{43} = \frac{z-2}{6}$$

$$\therefore \text{Line L contained in plane } 7x + py + qz + r = 0$$

$$35 + 43p + 6q = 0 \dots(i)$$

$$\text{and } 7(-3) + 5p + 2q + r = 0 \text{ or } 5p + 2q + r = 21$$

$$\dots(ii)$$

$$\text{and } 14 - p + q = 0 \dots(iii)$$

$$\Rightarrow p = 1, q = -13, r = 42$$

$$\therefore p + q + r = 30$$

6. Ans (60.00)

$$\sum_{r=0}^{10} (-1)^r 4^{10-r} {}^{30}C_r \cdot {}^{30-r}C_{10-r}$$

$$\Rightarrow \sum_{r=0}^{10} (-1)^r 4^{10-r} \frac{30!}{r!(30-r)!} \cdot \frac{(30-r)!}{20!(10-r)!}$$

$$\Rightarrow \sum_{r=0}^{10} (-1)^r \cdot 4^{10-r} \cdot {}^{30}C_{10} \cdot {}^{10}C_r = {}^{30}C_{10} \cdot 3^{10}$$

$$\text{Hence } k = 10$$

$$\text{Also, } \sum_{r=0}^{20} {}^{30}C_r {}^{20}C_r = \sum_{r=0}^{20} {}^{30}C_r {}^{20}C_{20-r} = {}^{50}C_{20}$$

$$\therefore n = 50$$

7. Ans (771.00)

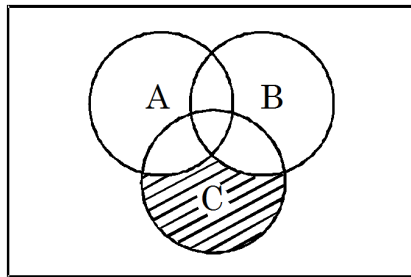
Let property, divisible by 5 = A $n(A) = 200$

divisible by 7 = B $n(B) = 142$

divisible by 3 = C $n(C) = 333$

$$n(\bar{A} \cap \bar{B} \cap C)$$

$$= n(A \cup B \cap \bar{C})$$



$$= n(C) - n(C \cap A \cup B)$$

$$= 333 - [n(C \cap A) + n(C \cap B) - n(C \cap A \cap B)]$$

$$= 333 - [66 + 47 - 9]$$

$$= 333 - 104$$

$$= 229$$

$$\text{Hence Probability} = \frac{229}{1000}$$

8. Ans (125.00)

$$T_r = \frac{r^2}{(2r-1)(2r+1)} = \frac{1}{4} \left(\frac{r}{2r-1} + \frac{r}{2r+1} \right)$$

$$4S = \sum_{r=1}^{500} \left(\frac{r}{2r-1} + \frac{r}{2r+1} \right)$$

$$= 1 + \left(\frac{1}{3} + \frac{2}{3} \right) + \left(\frac{2}{5} + \frac{3}{5} \right) + \left(\frac{3}{7} + \frac{4}{7} \right) + \dots + \frac{500}{999} + \frac{500}{1001}$$

$$\Rightarrow 4s = 1 + 499 + \frac{500}{1001} \Rightarrow [s] = 125$$

9. Ans (15.00)

$$\frac{dy}{dx} \sqrt{4-x^2} - \frac{x}{\sqrt{4-x^2}} y = 7x^6 + 2x$$

$$d \left(\sqrt{4-x^2} \cdot y \right) = (7x^6 + 2x) dx$$

$$\Rightarrow \sqrt{4-x^2} y = x^7 + x^2 + C$$

$$\therefore f(0) = 1 \Rightarrow c = 2$$

$$\therefore \int_{-\sqrt{3}}^{\sqrt{3}} \left(\frac{x^7}{\sqrt{4-x^2}} + \frac{x^2+2}{\sqrt{4-x^2}} \right) dx = 2 \int_0^{\sqrt{3}} \frac{x^2+2}{\sqrt{4-x^2}} dx$$

$$\text{put } x = 2\sin\theta$$

$$dx = 2 \cos \theta d\theta$$

$$= 2 \int_0^{\pi/3} \frac{4\sin^2\theta + 2}{2 \cos \theta} \cdot 2 \cos \theta d\theta$$

$$= 2 \int_0^{\pi/3} [4 - 2 \cos 2\theta] d\theta$$

$$= 2 \left[\frac{4\pi}{3} - \frac{\sqrt{3}}{2} \right] = \frac{8\pi}{3} - \sqrt{3}$$

$$\therefore a = \frac{8}{3}, b = -1$$

10. Ans (18.00)

Feet of the perpendicular (N_1 and N_2) from focus upon any tangent to parabola lies on the tangent line at the vertex.

Now equation of SN_1 is $x + y = \lambda$ passing through (1, 2)

$$\Rightarrow \lambda = 3$$

Equation of SN_1 is $x + y = 3$

Solving $x + y = 3$ and $y = x$,

$$\text{we get } N_1 \equiv \left(\frac{3}{2}, \frac{3}{2} \right)$$

||ly equation of SN_2 is $x - y = \mu$ passing through (1, 2)

$$\Rightarrow \mu = -1$$

Equation of SN_2 is $y - x = 1$

Solving $y - x = 1$ and $y = -x$,

$$\text{we get } N_2 \equiv \left(\frac{-1}{2}, \frac{1}{2} \right)$$

Now equation of tangent line at vertex is,
 $2x - 4y + 3 = 0$

Distance of $S(1, 2)$ from tangent at vertex is

$$= \frac{|2 - 8 + 3|}{\sqrt{20}} = \frac{3}{2\sqrt{5}}$$

$$= \frac{1}{4} \times \text{latus rectum and hence length of latus rectum} = \frac{6}{\sqrt{5}}, \text{ harmonic mean of AS and BS is}$$

$$\text{half the length of latus rectum} = \frac{3}{\sqrt{5}}$$

Hence $m = 3, n = 5$

$$\Rightarrow m + 3n = 18$$

PART-3 : MATHEMATICS
SECTION-III
1. Ans (3)

Given expression is

$$(f(x) f'(x) - x) ((f'(x))^2 + f(x) f''(x) - 1) = 0$$

$$\text{consider } g(x) = f^2(x) - x^2$$

since $g(x) = 0$ has 3 real roots

$\Rightarrow g'(x) = 0$ must have minimum 2 real roots

and $g''(x) = 0$ must have minimum 1 real root

2. Ans (3)

Let $f(x) = x^5 - x^3 + x - 2$ then $f(1) = -1$ and

$f(\sqrt{2}) = 3\sqrt{2} - 2$ implies that the real root lies in

$$(1, \sqrt{2})$$

$$\text{Now, } \alpha^5 - \alpha^3 + \alpha = \frac{\alpha(\alpha^6 + 1)}{\alpha^2 + 1}$$

$$\text{but } \alpha^5 - \alpha^3 + \alpha = 2 \quad \therefore \frac{\alpha(\alpha^6 + 1)}{\alpha^2 + 1} = 2$$

$$\Rightarrow \alpha^6 + 1 = \frac{2\alpha^2 + 2}{\alpha} = 2\alpha + \frac{2}{\alpha}$$

$$1 < \alpha < \sqrt{2} \quad \therefore \frac{1}{\sqrt{2}} < \frac{1}{\alpha} < 1$$

$$\therefore 1 + \frac{1}{\sqrt{2}} < \alpha + \frac{1}{\alpha} < 1 + \sqrt{2} \text{ but } \alpha + \frac{1}{\alpha} > 2$$

$$\therefore 2 < \alpha + \frac{1}{\alpha} < 1 + \sqrt{2} \therefore 4 < 2\alpha + \frac{2}{\alpha} < 2 + 2\sqrt{2}$$

$$\therefore 4 < \alpha^6 + 1 < 2 + 2\sqrt{2}$$

$$\Rightarrow 3 < \alpha^6 < 1 + 2\sqrt{2} \approx 3.8284 \quad \therefore [\alpha^6] = 3$$

3. Ans (1)

$$V = 24 [\vec{a} \vec{b} \vec{c}]^2$$

$$\text{Now } [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 1 + \sin \theta & \cos \theta & \sin 2\theta \\ \sin(\theta + \lambda) & \cos(\theta + \lambda) & \sin 2(\theta + \lambda) \\ \sin(\theta - \lambda) & \cos(\theta - \lambda) & \sin 2(\theta - \lambda) \end{vmatrix}$$

$$\text{where } \lambda = \frac{2\pi}{3}$$

$$R_1 \rightarrow R_1 + (R_2 + R_3)$$

$$[\vec{a} \vec{b} \vec{c}] = -\frac{\sqrt{3}}{2} \cos 3\theta$$

$$\therefore \cos^2 3\theta = 1 \Rightarrow \theta = \frac{\pi}{3}. \text{ Ans.}$$

4. Ans (4)

From given equation, we have

$$\{f(4x) - f(2x)\} - \{f(2x) - f(x)\} = x^2$$

$$\Rightarrow \left\{f(x) - f\left(\frac{x}{2}\right)\right\} - \left\{f\left(\frac{x}{2}\right) - f\left(\frac{x}{4}\right)\right\} = \left(\frac{x}{4}\right)^2$$

Replacing x by $\frac{x}{2}$ repeatedly $n - 1$ times and adding, we have

$$\left\{f(x) - f\left(\frac{x}{2^n}\right)\right\} - \left\{f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^{n+1}}\right)\right\} = \frac{x^2}{16} \cdot \frac{1 - \left(\frac{1}{4}\right)^n}{1 - \frac{1}{4}}$$

$$\therefore \lim_{n \rightarrow \infty} f\left(\frac{x}{2^n}\right) = \lim_{n \rightarrow \infty} f\left(\frac{x}{2^{n+1}}\right)$$

$$= f(0) \text{ \{As } f(x) \text{ is a continuous function\}}$$

$$\therefore f(x) - f(0) - f\left(\frac{x}{2}\right) + f(0) = \frac{x^2}{16} \cdot \frac{4}{3} = \frac{x^2}{12}$$

$$\Rightarrow f(x) - f\left(\frac{x}{2}\right) = \frac{x^2}{12}$$

Repeating the above procedure, we have

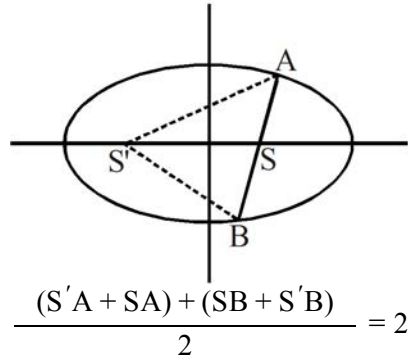
$$\lim_{n \rightarrow \infty} \left\{f(x) - f\left(\frac{x}{2^n}\right)\right\} = \frac{x^2}{12} \cdot \frac{1}{1 - \frac{1}{4}}$$

$$\Rightarrow f(x) - f(0) = \frac{x^2}{12} \cdot \frac{1}{1 - \frac{1}{4}} \Rightarrow f(x) - f(0) = \frac{x^2}{9}$$

$$\therefore f(x) = \frac{x^2}{9} + 1 \text{ \{As } f(0) = 1\} ;$$

$$\therefore f(3\sqrt{3}) = \frac{(3\sqrt{3})^2}{9} + 1 = 4$$

5. Ans (8)



6. Ans (7)

$$\begin{aligned} &= \int \underbrace{x \sin x}_I \underbrace{e^{e^x+x^3+x} (e^x + 3x^2 + 1)}_{II} dx \\ &+ \int e^{e^x+x^3+x} (\sin x + x \cos x) dx \\ &= x \sin x e^{e^x+x^3+x} - \int (x \cos x + \sin x) e^{e^x+x^3+x} dx \\ &= x \sin x \cdot e^{e^x+x^3+x} + c \end{aligned}$$

7. Ans (3)

$$(20 \sin x)^3 + (15 \cos x)^3 + 12^3 = 3.20 \sin x.15 \cos x.12$$

$$(20 \sin x + 15 \cos x + 12) \cdot \frac{1}{2} \cdot$$

$$[(20 \sin x - 15 \cos x)^2 + (15 \cos x - 12)^2 + (12 - 20 \sin x)^2] = 0$$

$$\Rightarrow 20 \sin x + 15 \cos x + 12 = 0 \text{ or } 20 \sin x = 15$$

$$\cos x = 12$$

Case-I : when $20 \sin x + 15 \cos x + 12 = 0$

$$\Rightarrow 20 \cdot \frac{2t}{1+t^2} + 15 \cdot \frac{1-t^2}{1+t^2} + 12 = 0$$

$$\text{where } t = \tan \frac{x}{2}$$

$$\Rightarrow 3t^2 - 40t - 27 = 0 \Rightarrow \tan \frac{x}{2} = \frac{40 \pm \sqrt{1924}}{6}$$

$$\text{As } 0 < x < 2\pi \quad \therefore 0 < \frac{x}{2} < \pi$$

$\therefore$ only one solution each in $(0, \pi/2)$ & $(\pi/2, \pi)$

Case-II : when $20 \sin x = 15 \cos x = 12$

$$\Rightarrow \sin x = 3/5 \text{ and } \cos x = 4/5$$

$\therefore$ only one solution in $(0, 2\pi)$

$\therefore$ total 3 distinct solutions in $(0, 2\pi)$

8. **Ans (3)**

Let given limit = L, then

$$L = \lim_{x \rightarrow 2} \frac{(1 + x + x^2 + x^3 + \dots + x^{n-1}) - (2^n - 1)}{x - 2}$$

Replacing $2^n - 1$ by $1 + 2 + 2^2 + \dots + 2^{n-1}$, we have

$$L = \lim_{x \rightarrow 2} \frac{(1 + x + x^2 + \dots + x^{n-1}) - (1 + 2 + 2^2 + \dots + 2^{n-1})}{x - 2}$$

$$L = \lim_{x \rightarrow 2} \frac{(x - 2) + (x^2 - 2^2) + (x^3 - 2^3) + \dots + (x^{n-1} - 2^{n-1})}{x - 2}$$

$$L = 1 + 2 + 2^2 + 2^3 + \dots + (n-1) \cdot 2^{n-2}$$

$$2L = 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + (n-2) \cdot 2^{n-2} + (n-1) \cdot 2^{n-1}$$

$$\therefore 2L - L = (n-1) \cdot 2^{n-1} - [1 + 2 + 2^2 + 2^3 + \dots + 2^{n-2}]$$

$$L = (n-1) \cdot 2^{n-1} - (2^{n-1} - 1) = (n-2) \cdot 2^{n-1} + 1$$

$$\text{Given that } L = 4097 \quad \therefore (n-2) \cdot 2^{n-1} + 1$$

$$= 4097 = 2^{12} + 1 \Rightarrow (n-2) \cdot 2^{n-1} = 2^{12}$$

$$\Rightarrow n-2 = 2^{13-n}$$

The value of n which satisfies above equation is

$$n = 10 \therefore [\sqrt{n}] = 3.$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

AIOT

13-05-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A,B,C,D	A,D	A,B	B,D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	D	C	A	D	B	B
SECTION-I (iii)	Q.	11	12	13	14		
	A.	A	B	D	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.67	5.56	70.00	200.00	400.00	3.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	A,B	C,D	A,C	A,B,C,D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C	B	A	B	B	D
SECTION-I (iii)	Q.	11	12	13	14		
	A.	B	B	C	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	39.40	35.70	8.00	7.10	5.00	13.25

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A,C	B,D	A	A,C,D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A	C	B	C	B	B
SECTION-I (iii)	Q.	11	12	13	14		
	A.	A	B	A	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.00	0.00	2.00	42.00	9.22 or 9.23	0.50

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

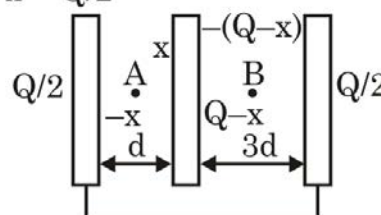
Field inside the conductor is zero.

2. Ans (A,D)

Since P.D between left and right plate is zero.

$$\frac{x}{A\epsilon_0}d = \frac{Q-x}{3A\epsilon_0} \times 3D$$

$$x = Q/2$$

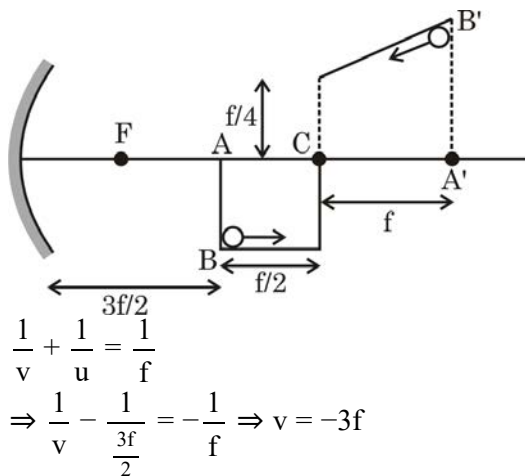


PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

The image of the object can be easily drawn by using the mirror formula. Right upright side of wire frame is on centre of curvature so its image would be real, inverted and of same size forming at the centre of curvature. For left upright side i.e. for point A. Use mirror formula.



i.e. image of A forms at $-3f$ distance from pole.

Height of image of AB can be computed by

using magnification formula $m = \frac{h_1}{h_0} = -\frac{v}{u}$

$$\Rightarrow h_i = - \left[\frac{-3f}{-\frac{3f}{2}} \right] \times \left[-\frac{f}{4} \right] = \frac{f}{2}$$

The top points of upright section of image of wire frame would be joined by a straight line, as the image of linear object would be always straight. The particle always remain in contact with the lower edge of wire frame so the image of it moves along the slant part of the image.

8. **Ans (D)**

It is clear from ray diagram that the velocity of image of particle is varying.

PART-1 : PHYSICS

SECTION-I (iii)

12. **Ans (B)**

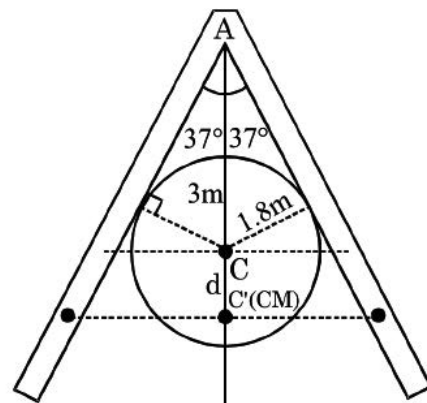
$$(A) \phi = 3 \int_0^L y^2 dy L t = \beta L^4 t \Rightarrow \varepsilon = \beta_0 L_4$$

$$(B) \phi = \beta_0 t \int_0^L z \int_0^L x dx dz = \frac{\beta_0 L^4 t}{4} \Rightarrow \varepsilon = \frac{\beta_0 L^4}{4}$$

$$(C) \varepsilon = \frac{5\beta_0 L^4}{4}$$

$$(D) \varepsilon = \frac{5\beta_0 L^4}{4}$$

13. **Ans (D)**



$$x_{CM} = \frac{\int x dm}{\int dm} = \frac{\int_0^\ell x \cdot 15x^2 dx}{\int_0^\ell 15x^2 dx} = \frac{3\ell}{4} \quad \dots(i)$$

For static equilibrium CM of the rods must lie below the centre of the disc.

$$\frac{3\ell_{\min}}{4} \cos 37^\circ = 3$$

$$\ell_{\min} = 5m \quad \dots(ii)$$

MI of the rods about O.

$$2 \int dm x^2 = 2 \int_0^\ell x^2 \times 15x^2 dx$$

$$= 2 \times 15 \times \frac{\ell^5}{5} = 6\ell^5$$

Mass of the rods,

$$m = \int dm = 2 \int_0^\ell 15x^2 dx$$

$$m = \int dm = 2 \int_0^{\ell} 15x^2 dx = 10\ell^3 \dots(iii)$$

$$OC' = \frac{3\ell}{4} \times \cos 37^\circ = \frac{3\ell}{5} \dots(iv)$$

$$d = CC' = \frac{3\ell - 15}{5} \dots(v)$$

MI of the rods about C.

$$I_C = I_O - m(OC')^2 + md^2$$

$$= 6\ell^5 - 36\ell^4 + 90\ell^3 \dots(vi)$$

Time period for a small oscillation

$$T = 2\pi \sqrt{\frac{I_C}{mgd}} = 2\pi \sqrt{\frac{\ell^2 - 6\ell + 15}{10(\ell - 5)}}$$

$$\text{For } \ell = 10 \text{ m, } T = 2\pi \sqrt{\frac{11}{10}}$$

$$\text{For } \ell = 8 \text{ m, } T = 2\pi \sqrt{\frac{31}{30}}$$

14. Ans (B)

$$\vec{V}_{P,P} = V_2 \hat{i} + 25 \hat{j} + V_1 \hat{k}$$

$$\vec{a}_{P,P} = -2 \hat{i} - 12.5 \hat{j}$$

$\vec{V}_{P,P}$ = Velocity of particle relative to platform

$$\text{Time} = \frac{2 \times 25}{12.5} = 4 \text{ sec}$$

$$8 \leq V_2 \times 4 - 1/2 \times 2 \times 4^2 \leq 16$$

$$6 \leq V_2 \leq 8$$

$$16 \leq V_1 \times 4 \leq 24$$

$$4 \leq V_1 \leq 6$$

$$Y = 25 \times 4 - \frac{1}{2} \times 10 \times 4^2$$

$$= 100 - 80 = 20 \text{ m}$$

PART-1 : PHYSICS

SECTION-II

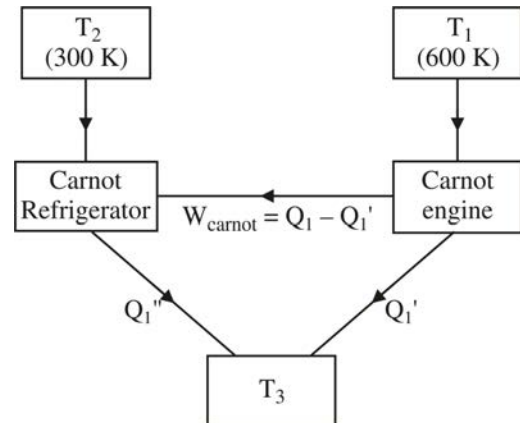
3. Ans (70.00)

$$v = \frac{F}{m} \times t$$

4. Ans (200.00)

$$I^2 R \times t = ms \Delta T$$

6. Ans (3.00)



$$\eta_{\text{carnot engine}} = \frac{Q_1 - Q_1'}{Q_1} = \frac{T_1 - T_3}{T_1}$$

$$= \frac{\text{Work of Carnot engine}}{\text{Heat supplied to the Carnot engine}} = \frac{W_{\text{carnot}}}{Q_1}$$

$$\text{or } W_{\text{carnot}} = Q_1 \left(\frac{T_1 - T_3}{T_1} \right) \dots(i)$$

Also,

$$\text{COP}_{(\text{carnot refrigerator})} = \frac{Q_2}{Q_1'' - Q_2} = \frac{T_2}{T_3 - T_2}$$

$$= \frac{\text{Heat absorbed}}{W_{\text{carnot}}} = \frac{Q_2}{W_{\text{carnot}}}$$

$$\text{or } W_{\text{carnot}} = Q_2 \left(\frac{T_3 - T_2}{T_2} \right) \dots(ii)$$

(i) Temperature, T_3 :

From equations (i) and (ii), we get

$$Q_1 \left(\frac{T_1 - T_3}{T_1} \right) = Q_2 \left(\frac{T_3 - T_2}{T_2} \right)$$

$$\therefore \frac{Q_2}{Q_1} = \frac{T_2}{T_1} \left(\frac{T_1 - T_3}{T_3 - T_2} \right)$$

$$\text{or } \frac{Q_2}{Q_1} = 1 = \frac{300}{600} \left(\frac{600 - T_3}{T_3 - 300} \right)$$

$$\text{or } 600 - T_3 = 2(T_3 - 300)$$

$$600 - T_3 = 2T_3 - 600$$

$$\text{or } T_3 = 400 \text{ K}$$

Hence, temperature, $T_3 = 400 \text{ K}$

(ii) Efficiency of Carnot engine,

$$\eta_{\text{carnot engine}} = \frac{T_1 - T_3}{T_1}$$

$$= \frac{600 - 400}{600} = 0.3333 = 33.33\%$$

$$\text{COP}_{\text{refrigerator}} = \frac{T_2}{T_3 - T_2} = \frac{300}{400 - 300} = 3$$

PART-2 : CHEMISTRY

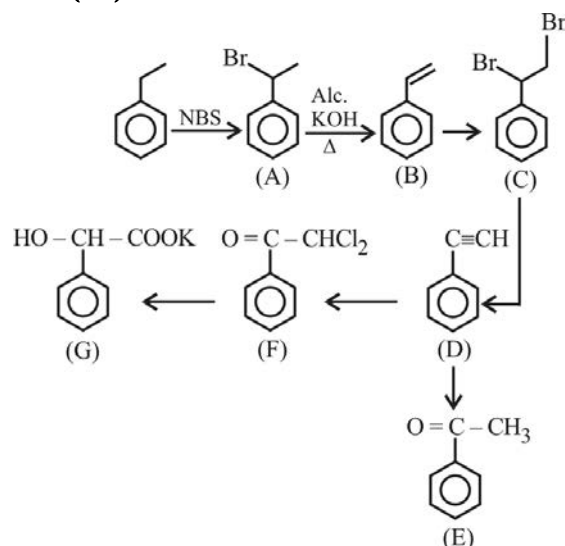
SECTION-I (i)

- Ans (A,B)**
Spontaneity is not related with resistivity.
- Ans (C,D)**
(A) Opposite is given.
(B) EMF is intensive property.
- Ans (A,B,C,D)**
X = NaN₃ Y = NaSCN Z = NaCN
R = [Fe(SCN)]⁺² B = Na^{III}Fe^{II}[Fe(CN)₆]

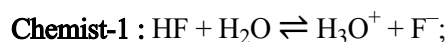
PART-2 : CHEMISTRY

SECTION-I (ii)

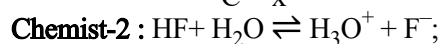
- Ans (B)**



- Ans (B)**



$$K_1 = \frac{x \cdot x}{C - x}$$



$$K_1 = \frac{x \cdot (x - y)}{C - x - y}$$

$$\text{HF} + \text{F}^- \rightleftharpoons \text{HF}_2^-$$

$$K_2 = \frac{y}{(C - x - y)(x - y)}$$

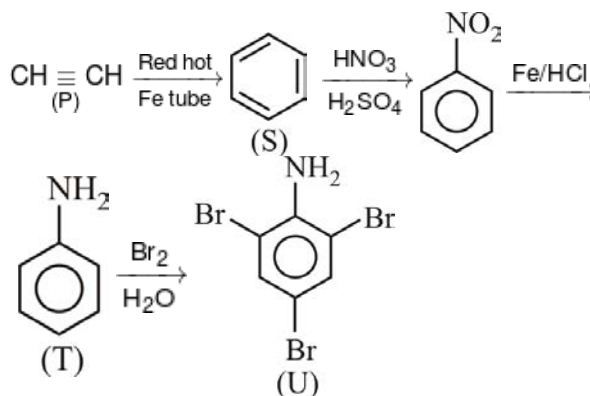
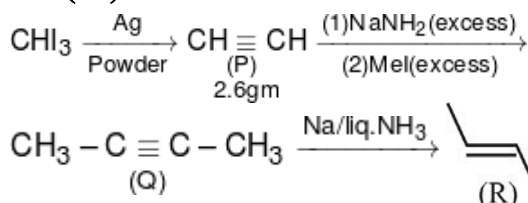
From question : $\frac{x \cdot x}{C - x} = \frac{x \cdot (x - y)}{C - x - y} \Rightarrow C = 2x$

and $K_1 = x$

PART-2 : CHEMISTRY

SECTION-I (iii)

- Ans (B)**



$$\text{Moles of CHI}_3 = 2x \text{ moles of P} = \frac{2.6}{26} \times 2 = 0.2$$

$$x = 78.8 \text{ gm}$$

$$n = 4$$

$$\text{Moles of (S)} = \frac{1}{3} \times \text{moles of (P)}$$

$$\text{Moles of (T)} = \frac{1}{3} \times 0.1 \times 0.9$$

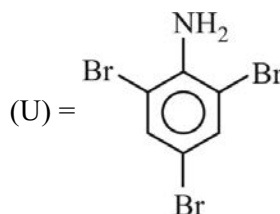
$$\text{Moles of (U)} = \frac{1}{3} \times 0.1 \times 0.9 \times .8 = 0.024$$

$$\text{Moles of (P)} = \text{Moles of Q} = \text{Moles of (R)} = 0.1$$

$$\text{R} = \text{C}_4\text{H}_8 \quad M_R = 56$$

$$W_{\text{C}_4\text{H}_8} = 56 \times 0.1 = 5.6 \text{ gm}$$

$$y = 5.6$$



$$\text{C}_6\text{H}_4\text{NBr}_3$$

$$M = 330$$

$$W_U = 330 \times .024 = 7.92 \text{ gm}$$

$$z = 7.92$$

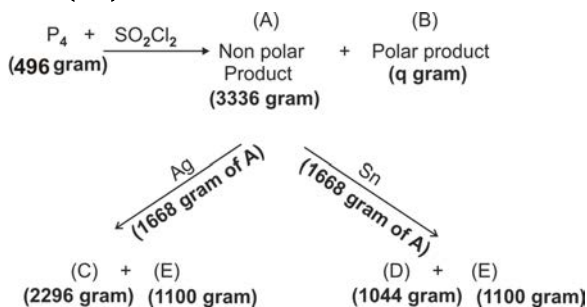
12. **Ans (B)**

$$\ell = 1 \text{ and } n - \ell - 1 = 1 \Rightarrow n = 3$$

Angular function depends only on Q, hence orbital is $3p_z$.

$$\text{for radial node, } \sigma = 6 \Rightarrow r = 4.5 a_0$$

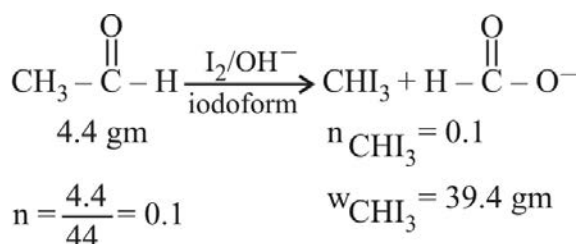
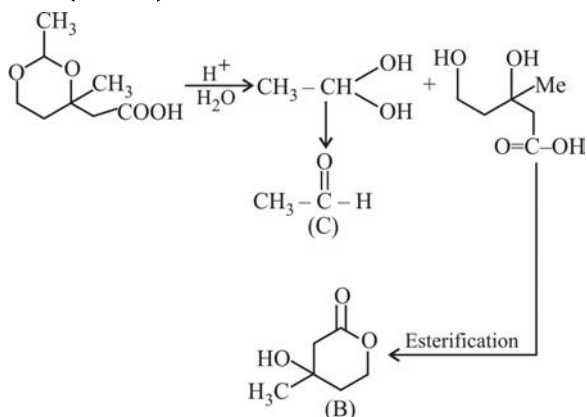
14. **Ans (D)**



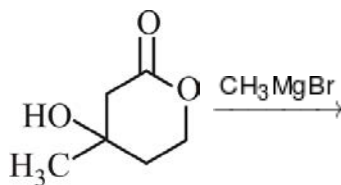
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (39.40)**



2. **Ans (35.70)**

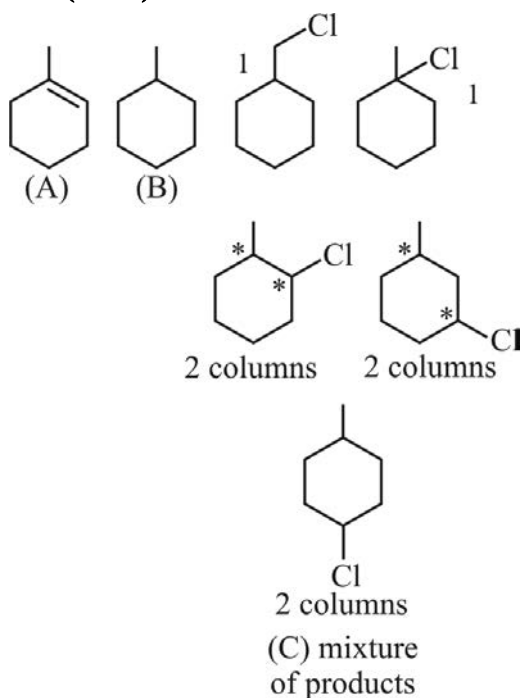


moles of CH_3MgBr used per mole of reactant = 3

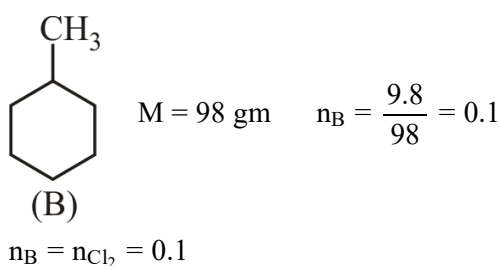
Total moles used $CH_3MgBr = 0.3$

Mass of $CH_3MgBr = 0.3 \times 119 = 35.70 \text{ gm}$

3. **Ans (8.00)**



4. **Ans (7.10)**



$$w_{Cl_2} = 0.1 \times 71 = 7.10 \text{ gm}$$

5. **Ans (5.00)**

For 1 Cd-atom, there should be 1 O-atom but the actual number is 0.95. Hence, % of anion sites

$$\text{vacant} = \frac{1 - 0.95}{1} \times 100 = 5\%$$

6. **Ans (13.25)**

$$d = \frac{m}{V} = \frac{(4 \times 112 + 4 \times 0.95 \times 16)}{6 \times 10^{23} \times (400 \times 10^{-10})^3}$$

$$= 13.25 \text{ gm/cm}^3$$

PART-3 : MATHEMATICS

SECTION-I (i)

 1. **Ans (A,C)**

$$f(x) = \frac{1}{1-x}; x \neq 1$$

$$f(f(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}$$

$$f(f(f(x))) = \frac{1}{1 - \left(\frac{x-1}{x}\right)} = x$$

 $\therefore h(x)$ be a polynomial $g(x) = x$

 for this $n = 3, 6, 9, 12 \dots$

Hence option A and C

 Equation of tangent at $(0, -1)$ is

$$y + 1 = 0(x - 0)$$

 $\therefore \{ \text{where } h'(x) = 6x^5 + 3x^2 \text{ \& } h'(0) = 0 \}$

$$y = -1$$

$$\text{Required area} = \int_{-1}^0 (-1 - h(x)) dx$$

$$\int_{-1}^0 (-1 - x^6 - x^3 + 1) dx$$

$$\Rightarrow \left[-\frac{x^7}{7} - \frac{x^4}{4} \right]_{-1}^0 = \frac{3}{28}$$

 2. **Ans (B,D)**

$$f(x) = \frac{1}{1-x}; x \neq 1$$

$$f(f(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}$$

$$f(f(f(x))) = \frac{1}{1 - \left(\frac{x-1}{x}\right)} = x$$

 $\therefore h(x)$ be a polynomial $g(x) = x$

 for this $n = 3, 6, 9, 12 \dots$

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 Equation of tangent at $(0, -1)$ is

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$$\int_{-1}^0 (-1 - x^6 - x^3 + 1) dx$$

$$\Rightarrow \left[-\frac{x^7}{7} - \frac{x^4}{4} \right]_{-1}^0 = \frac{3}{28}$$

 3. **Ans (A)**

$$f(x) = f(2-x)$$

 $\Rightarrow f$ is symmetric about $x = 1$

$$\text{Now } -\frac{b}{2(-1)} = 1 \Rightarrow b = 2$$

$$\Rightarrow f(x) = -x^2 + 2x + c$$

$$\alpha + \beta = \frac{-2}{-1} = 2$$

$$\alpha\beta = -c$$

$$\alpha(2-\alpha) = -c$$

$$c = \alpha^2 - 2\alpha$$

$$\therefore c \in (0, 1)$$

$$\text{Now } \alpha^2 - 2\alpha - c = 0$$

$$\alpha\beta = \frac{2 \pm \sqrt{4+4c}}{2} = 1 \pm \sqrt{1+c}$$

$$\text{Let } \alpha = 1 + \sqrt{1+c}, \beta = 1 - \sqrt{1+c}$$

$$\text{Now } \alpha \in (2, 1 + \sqrt{2}) \text{ \& } \beta \in (1 - \sqrt{2}, 0)$$

$$[\alpha] = 2, [\beta] = -1$$

$$\max([\alpha], [\beta]) = 2$$

Hence option (A)

4. Ans (A,C,D)

Now $c \in (-1, 0)$

$$-1 < a^2 - 2a < 0$$

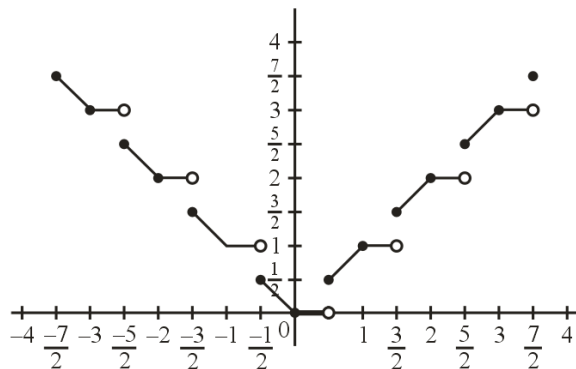
$$\Rightarrow a \in (0, 1) \cup (1, 2)$$

Now verify options

PART-3 : MATHEMATICS

SECTION-I (ii)

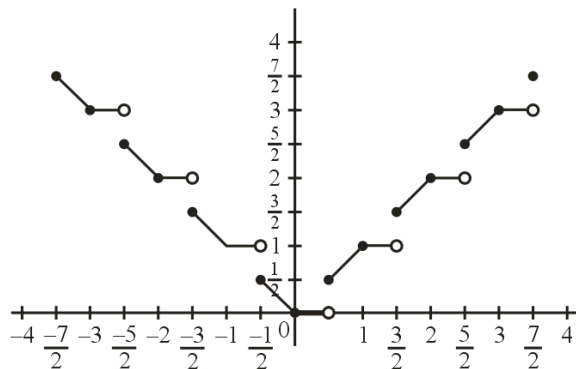
5. Ans (A)



$$L = 7, M = 14$$

$$\therefore L + M = 21$$

6. Ans (C)



$$L = 7, M = 14$$

$$\therefore L + M = 21$$

7. Ans (B)

$$f(x) = a^2x^3 + ax^2 - x + b$$

$$f'(x) = 3a^2x^2 + 2ax - 1$$

$$\text{Local minimum } x = -1$$

$$f'(-1) = 3a^2 - 2a - 1 = 0$$

$$(a - 1)(3a + 1) = 0$$

$$\therefore a = 1, a = -\frac{1}{3}$$

$$\text{Case-1 : } a = 1$$

$$f(x) = x^3 + x^2 - x + b$$

$$f'(-1) = -1 + 1 + 1 + b > 0$$

$$\boxed{b > -1} \quad x = \{0, 1, 2, \dots\}$$

$$\text{Case-2 : } a = -\frac{1}{3}$$

$$f(x) = \frac{1}{9}x^3 - \frac{1}{3}x^2 - x + b$$

$$f(-1) = -\frac{1}{9} - \frac{1}{3} + 1 + b > 0$$

$$b > -\frac{5}{9}$$

$$y = \{0, 1, 2, \dots\}$$

Now

$$\sum_{n \in x} \left(\frac{\cos \theta}{2} \right)^n = 1 + \frac{\cos \theta}{2} + \left(\frac{\cos \theta}{2} \right)^2 + \dots \infty = \frac{2}{2 - \cos \theta}$$

$$\sum_{n \in y} \left(\frac{\cos \frac{\theta}{2}}{3} \right)^n = \frac{3}{3 - \cos \frac{\theta}{2}}$$

$$Z = \frac{2}{2 - \cos \theta} + \frac{3}{3 - \cos \frac{\theta}{2}} \quad Z_{\max}$$

For $\theta = 0$

$$2 + \frac{3}{2} = \frac{7}{2}$$

8. Ans (C)

$$a \in I \Rightarrow a = 1, b = 1$$

$$f(x) = x^3 + x^2 - x + 1$$

$$f(0) = 1$$

$$f(f(0)) = f(1) = 2$$

$$T_{(0,1)} : y - 1 = f'(0)(x - 0)$$

$$y - 1 = -1(x) \therefore x + y = 1$$

$$T_{(1,2)} : y - 2 = 4(x - 1)$$

$$y = 4x - 2$$

$$\therefore \text{point of intersection } \left(\frac{3}{5}, \frac{2}{5} \right)$$

9. Ans (B)

Putting $x^2 = 4y$ in equation of given circle

we get $4y + y^2 - 10y + 25 - r^2 = 0$

$$\Rightarrow y^2 - 6y + 25 - r^2 = 0$$

As parabola is tangent $\therefore D = 0$

$$36 - 4(25 - r^2) = 0 \therefore r = 4$$

Let $y = mx + c$ is tangent to $x^2 = 4y$

$$x^2 = 4(mx + c)$$

$$\therefore x^2 - 4mx - 4c = 0 \therefore D = 0 \text{ we get } m^2 = -c$$

$$\therefore y = mx - m^2 \text{ tangent to } x^2 = 4y$$

Above tangent is also tangent to circle

$$\text{we get } \frac{(5 + m^2)}{\sqrt{1 + m^2}} = 4 \therefore m = \pm\sqrt{3}$$

point of contact $P(2m_1, m_1^2)$; $Q(2m_2, m_2^2)$

$$\therefore P(2\sqrt{3}, 3) ; Q(-2\sqrt{3}, 3)$$

Equation of common tangent is $y = \pm\sqrt{3}x - 3$

and will intersect at $R(0, -3)$

centre of circle is $S(0, 5)$ and chord is $y = 3$

equation of latus rectum $\Rightarrow y = 1$

equation of directrix $\Rightarrow y = -1$

$$SM = 5 - \frac{1}{3} = \frac{14}{3}$$

$$\text{centroid} = \left(0, -\frac{7}{6}\right)$$

$$\text{area } \Delta PQR = \frac{20}{3\sqrt{3}}$$

10. Ans (B)

tangent at $(-4, 4)$

$$xx_1 = 4 \left(\frac{y + y_1}{2} \right)$$

$$-4x = 4 \left(\frac{y + 4}{2} \right)$$

$$2x + y + 4 = 0$$

$$(x + 4)^2 + (y - 4)^2 + \lambda(2x + y + 4) = 0$$

$$16 + 16 + \lambda(4) = 0 \therefore \lambda = -8$$

$$x^2 + y^2 - 8x - 16y = 0$$

PART-3 : MATHEMATICS

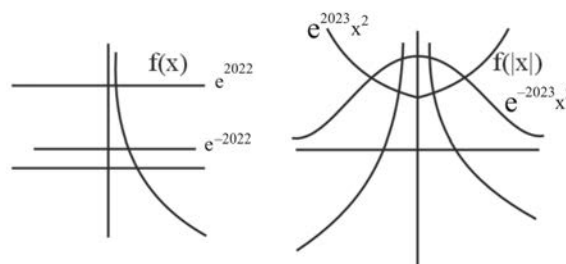
SECTION-I (iii)

11. Ans (A)

$$\lim_{n \rightarrow \infty} n \left(x^{\frac{2022}{n}} - x^{\frac{2023}{n}} \right)$$

$$f(x) \lim_{t \rightarrow 0} \frac{x^{2022t} - x^{2023t}}{t} = 2022 \ln x - 2023 \ln x$$

$$f(x) = -\ln x$$



12. Ans (B)

$$a_{ij} = \tan^{-1}(\tan(i + j))$$

$$a_{11} = \tan^{-1}(\tan 2) = 2 - \pi$$

$$a_{22} = \tan^{-1}(\tan 4) = 4 - \pi$$

$$a_{33} = \tan^{-1}(\tan 6) = 6 - 2\pi$$

$$a_{12} = \tan^{-1}(\tan 3) = a_{21} = 3 - \pi$$

$$a_{13} = \tan^{-1}(\tan 4) = a_{31} = 4 - \pi$$

$$a_{23} = \tan^{-1}(\tan 5) = a_{32} = 5 - 2\pi$$

$$A = \begin{bmatrix} 2 - \pi & 3 - \pi & 4 - \pi \\ 3 - \pi & 4 - \pi & 5 - 2\pi \\ 4 - \pi & 5 - 2\pi & 6 - 2\pi \end{bmatrix}$$

$$\text{Tr}(A) = 12 - 4\pi$$

$$|A| = \begin{vmatrix} 2 - \pi & 3 - \pi & 4 - \pi \\ 1 & 1 & 1 - \pi \\ 1 & 1 - \pi & 1 \end{vmatrix} = \pi^3 - 2\pi^2 - 3\pi$$

$$= \begin{vmatrix} 2 - \pi & 1 & 2 \\ 1 & 0 & -\pi \\ 1 & -\pi & 0 \end{vmatrix}$$

$$1(-\pi - 0) + \pi(-\pi(2 - \pi) - 2)$$

$$\Rightarrow -\pi + \pi(-2\pi + \pi^2 - 2)$$

$$\Rightarrow -\pi - 2\pi^2 + \pi^3 - 2\pi \Rightarrow \pi^3 - 2\pi^2 - 3\pi$$

13. Ans (A)

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} \tan\left(\frac{x}{\sqrt{1+x^2}}\right) + 2\sqrt{1-x^2} \sin \frac{x}{\sqrt{1-x^2}} - 3x}{x^p}$$

using expansion

$$P = 5 \text{ and } M = -\frac{31}{60}$$

$$L = \frac{n^3 \cdot \sum_{r=1}^n e^{\frac{r}{n}}}{(n+1)^m \cdot \sum_{r=1}^n r^{2m}} = \frac{n^4 \cdot \sum_{r=1}^n e^{\frac{r}{n}} \cdot \frac{1}{n}}{(n+1)^m \cdot n^{2m+1} \cdot \sum_{r=1}^n \left(\frac{r}{n}\right)^{2m} \cdot \frac{1}{n}}$$

for L to be finite $m = 1$ and

$$\Rightarrow L = \frac{\int_0^1 e^x dx}{\int_0^1 x^2 dx} = \frac{e-1}{\frac{1}{3}} = 3(e-1)$$

14. Ans (A)

$$(a) I = \frac{1}{2} \int_0^\infty \frac{\sin(a+b)x + \sin(a-b)x}{x} dx$$

$$= \frac{1}{2} \left[\int_0^\infty \frac{\sin(a+b)x}{x} dx + \int_0^\infty \frac{\sin(a-b)x}{x} dx \right] = \frac{1}{2} (\beta + \beta) = \beta$$

Ans. (R)

$$(b) \text{ Put } \sqrt{x} = t \quad 2 \int_0^\infty e^{-t^2} dt = \sqrt{x} \text{ Ans. (Q)}$$

$$(c) \int_0^\infty x^2 e^{-x^2} dx = \int_0^\infty \underbrace{x}_I \cdot \underbrace{x e^{-x^2}}_{II} dx = -x \frac{e^{-x^2}}{2} \Big|_0^\infty$$

$$+ \frac{1}{2} \int_0^\infty e^{-x^2} dx = \frac{\alpha}{2} \text{ Ans. (T)}$$

$$(d) \int_0^\infty \frac{\sin^2 x}{x^2} dx = \int_0^\infty \underbrace{\frac{1}{x^2}}_{II} \cdot \underbrace{\sin^2 x}_I dx = -\frac{\sin^2 x}{x} \Big|_0^\infty + \int_0^\infty \frac{\sin 2x}{x} dx$$

Put $2x = t$

$$= 0 + \int_0^\infty \frac{\sin t}{t} dt = \beta \quad \text{Ans. (R)}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$z = z_0 \cos^4 t + 2z_1 \cos^2 t \sin^2 t + z_2 \sin^4 t \quad (t \in \mathbb{R})$$

$$x + iy = a \cos^4 t + 2 \left(\frac{1}{2} + bi \right) \cos^2 t \sin^2 t + (1 + ci) \sin^4 t$$

$$x + iy = \sin^2 t + i(a \cos^4 t + 2b \cos^2 t \sin^2 t + c \sin^4 t)$$

$$x = \sin^2 t; y = a \cos^4 t + 2b \cos^2 t \sin^2 t + c \sin^4 t$$

where $0 \leq x \leq 1$

$$y = a(1-x)^2 + 2b(1-x)x + cx^2$$

$$y = (a-2b+c)x^2 - 2(a-b)x + a$$

$\therefore \omega_2$ curve is

$$y = -2(a-b)x + a$$

$$z = t + ibt^2$$

$$x + iy = t + ibt^2$$

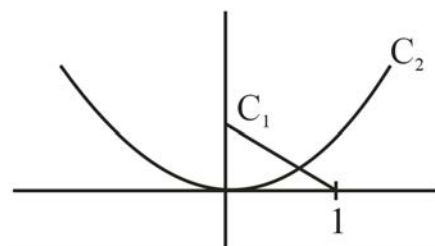
$$x = t, y = bt^2 \quad \therefore y = bx^2$$

Now

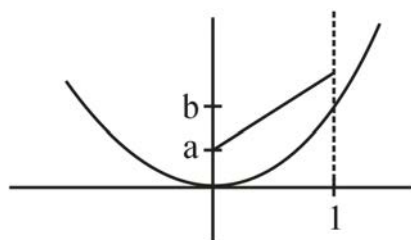
$$C_1 : y = -2(a-b)x + a; 0 \leq x \leq 1$$

$$C_2 : y = bx^2$$

for $b < a$ C_1 and C_2 will be of term



Case-II : $b > a$



2. Ans (0.00)

$$z = z_0 \cos^4 t + 2z_1 \cos^2 t \sin^2 t + z_2 \sin^4 t \quad (t \in \mathbb{R})$$

$$x + iy = a \cos^4 t + 2 \left(\frac{1}{2} + bi \right) \cos^2 t \sin^2 t + (1 + ci) \sin^4 t$$

$$x + iy = \sin^2 t + i(a \cos^4 t + 2b \cos^2 t \sin^2 t + c \sin^4 t)$$

$$x = \sin^2 t; y = a \cos^4 t + 2b \cos^2 t \sin^2 t + c \sin^4 t$$

where $0 \leq x \leq 1$

$$y = a(1-x)^2 + 2b(1-x)x + cx^2$$

$$y = (a - 2b + c)x^2 - 2(a - b)x + a$$

$\therefore \omega_2$ curve is

$$y = -2(a - b)x + a$$

$$z = t + ibt^2$$

$$x + iy = t + ibt^2$$

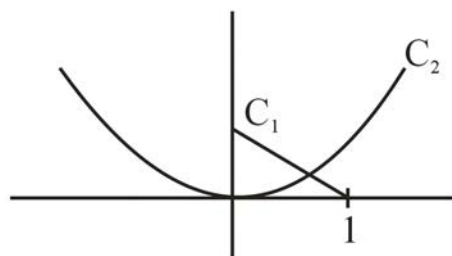
$$x = t, y = bt^2 \quad \therefore y = bx^2$$

Now

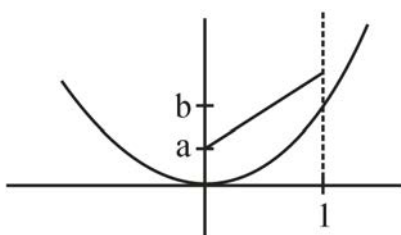
$$C_1 : y = -2(a - b)x + a; 0 \leq x \leq 1$$

$$C_2 : y = bx^2$$

for $b < a$ C_1 and C_2 will be of term



Case-II : $b > a$



3. Ans (2.00)

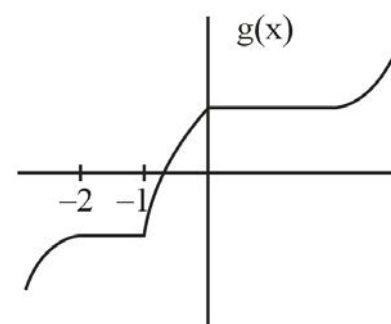
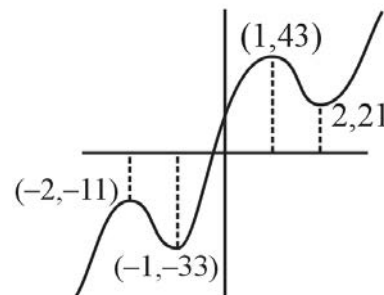
$$f(x) = 3x^5 - 25x^3 + 60x + 5$$

$$f'(x) = 15x^4 - 75x^2 + 60$$

$$= 15 [x^4 - 5x^2 + 4]$$

$$= 15 [x^2 - 4] [x^2 - 1]$$

$$= 15(x - 2)(x + 2)(x - 1)(x + 1)$$



4. Ans (42.00)

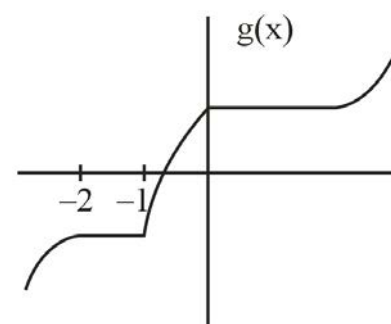
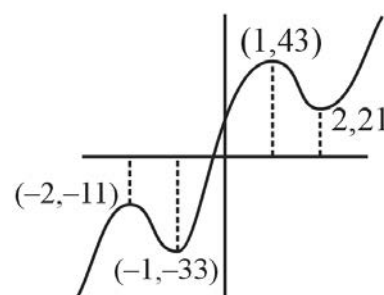
$$f(x) = 3x^5 - 25x^3 + 60x + 5$$

$$f'(x) = 15x^4 - 75x^2 + 60$$

$$= 15 [x^4 - 5x^2 + 4]$$

$$= 15 [x^2 - 4] [x^2 - 1]$$

$$= 15(x - 2)(x + 2)(x - 1)(x + 1)$$



5. Ans (9.22 or 9.23)

$$|P - nI| = \begin{vmatrix} 1-n & 2 & 0 \\ 2 & 1-n & 0 \\ 0 & 0 & 1-n \end{vmatrix} = 0$$

$$(1-n)((1-n)^2) - 2(2(1-n)) + 0 = 0$$

$$(1-n)((1-n)^2 - 4) = 0$$

$$(1-n)(1+n^2-2n-4) = 0$$

$$(1-n)(n^2-2n-3) = 0$$

$$(1-n)(n-3)(n+1) = 0$$

$$\therefore n = -1, n = 1, n = 3$$

$$n_1 = -1, n_2 = 1, n_3 = 3$$

$$PQP^T = N$$

$$Q = P^{-1}N(P^T)^{-1}$$

$$Q = P^{-1}N(P^{-1})^T$$

$$P^{-1} = \frac{\text{adj}P}{|P|} = \frac{1}{-3} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$Q = -\frac{1}{3} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \cdot -\frac{1}{3} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} -1 & -2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$Q = \frac{1}{9} \begin{bmatrix} 3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 27 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$Q^2 = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\text{tr}(Q^2) = \frac{2}{9} + 9 = \frac{83}{9} = 9.22 \text{ or } 9.23$$

using multiplication of matrix

$$\text{tr}(P) = 3 \quad \text{tr}(P^3) = 3^3 \quad \text{tr}(P^5) = 3^5 \dots$$

$$\text{tr}(P^2) = 3^2 + 2 \quad \text{tr}(P^4) = 3^4 + 2 \dots$$

6. Ans (0.50)

$$|P - nI| = \begin{vmatrix} 1-n & 2 & 0 \\ 2 & 1-n & 0 \\ 0 & 0 & 1-n \end{vmatrix} = 0$$

$$(1-n)((1-n)^2) - 2(2(1-n)) + 0 = 0$$

$$(1-n)((1-n)^2 - 4) = 0$$

$$(1-n)(1+n^2-2n-4) = 0$$

$$(1-n)(n^2-2n-3) = 0$$

$$(1-n)(n-3)(n+1) = 0$$

$$\therefore n = -1, n = 1, n = 3$$

$$n_1 = -1, n_2 = 1, n_3 = 3$$

$$PQP^T = N$$

$$Q = P^{-1}N(P^T)^{-1}$$

$$Q = P^{-1}N(P^{-1})^T$$

$$P^{-1} = \frac{\text{adj}P}{|P|} = \frac{1}{-3} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$Q = -\frac{1}{3} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \cdot -\frac{1}{3} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} -1 & -2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$Q = \frac{1}{9} \begin{bmatrix} 3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 27 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$Q^2 = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\text{tr}(Q^2) = \frac{2}{9} + 9 = \frac{83}{9} = 9.22 \text{ or } 9.23$$

using multiplication of matrix

$$\text{tr}(P) = 3 \quad \text{tr}(P^3) = 3^3 \quad \text{tr}(P^5) = 3^5 \dots$$

$$\text{tr}(P^2) = 3^2 + 2 \quad \text{tr}(P^4) = 3^4 + 2 \dots$$